

1-D:

D) According to the “**equi-marginal principle**” (also known as the law of equi-marginal utility), utility maximization occurs when a consumer allocates their income so that the **marginal utility per dollar spent** on each good is equal. This ensures that the consumer cannot reallocate spending to gain more total utility. If $MU_x/P_x = MU_y/P_y$ for goods 1 and 2, the consumer achieves maximum total utility given their budget constraint. Hence, D correctly reflects this core principle of consumer choice theory.

A is incorrect because equal total utility across commodities is not required.

B is incorrect as the total utility per dollar does not need to be the same; instead the marginal utility per dollar spent on both goods should be equal.

C is misleading as it refers to equal marginal utility from the last unit for both the goods.

2-B:

Consumers always face **budget constraints** that limit their choices, and these constraints are represented by the **budget line**. Indifference curves show combinations of goods that yield equal satisfaction, but the budget line limits what is affordable. **Hence, limited income is a crucial determinant which limits the consumer.**

A assumes complete preference ranking, which relies on full information and rationality — not always realistic.

C assumes perfect rationality, yet behavioural economics shows that consumers often act irrationally or inconsistently.

D suggests consumer will always get positive utility from consumption of more of the goods, but **consumption of additional units may lead to the marginal utility to diminish.**

3-D:

Correct answer: D — Unit cost falls as output increases; Unit cost rises as output increases

Diagram 1 shows a typical downward-sloping LRAC curve, representing **increasing returns to scale (economies of scale)** — as output rises, average costs fall due to increased efficiency.

Diagram 2 slopes upward, representing **diseconomies of scale (falling/decreasing returns to scale)**, where coordination and administrative inefficiencies cause costs to rise with output.

A- The option for Diagram 1 refers to “falling returns to scale,” which is incorrect because **falling returns to scale** actually causes **rising** average costs. Similarly, if MC was less than AC in Diagram 2, AC should fall. This is incorrect because in diagram 2, AC is seen to rise.

B- B incorrectly pairs falling and rising returns to scale — diagram 1 depicts increasing returns; similarly, diagram 2 depicts falling returns to scale

C- Diagram 2 here does not show falling Average fixed costs; instead, it shows rising AC.

4-A:

A) The compulsory use of helmets is a **consumption activity with positive externalities**. By wearing helmets, motorcyclists reduce the severity and frequency of head injuries in accidents. When motor-cyclists do not burdenize the healthcare system with their accidents, this benefits **third parties** by lowering the burden on emergency and hospital services. Hence, A captures the **external benefit to society** resulting from private consumption.

B is a **cost** incurred by the government to enforce the law — not an external benefit.

C (increased life expectancy for motorcycle users) is a **private benefit** to motorbike users — it reflects personal gain, not an external benefit.

D reflects **producer profits**, not a third-party benefit. The focus of external benefits is to benefit the third parties, who are not directly involved in the transaction.

5-D: To find out the concentration ratio of the four largest firms, we need to add the market shares of them:

From 2018 to 2020, the **market shares of the top four firms** all declined: Norton, Souton, Eastern, and Western each lost market share. Since the **concentration ratio** is the total market share held by the largest firms, a fall in all top four implies the **four-firm concentration ratio declined**, indicating **reduced market concentration** and more competition.

5 The table shows the market shares of firms selling a very similar product in a country between 2018 and 2020.

firm	2018	2020
Norton	28.4	27.4
Souton	18.9	17.6
Eastern	17.1	14.7
Western	12.5	11.4
Scotsdale	7.2	8.5
Welshpool	5.8	8.2
Greenland	3.9	4.8
Franklin	2.2	3.6
Bellweather	2.1	1.9
others	1.9	1.9

What can be concluded from the data between the two years?

- A Market concentration increased.
- B Sales of the top four firms fell.
- C Scotsdale and Welshpool's profits rose.
- D The concentration ratio of the four largest firms fell.

4 largest firms are

1) Norton

2) Souton

3) Eastern

4) Western

2018 : concentration ratio $\Rightarrow$

$$28.4 + 18.9 + 17.1 + 12.5 = 76.9$$

2020 : $\Rightarrow$

$$27.4 + 17.6 + 14.7 + 11.4 = 71.1$$

concentration ratio of 4 largest firms fell.

A is incorrect because market concentration **decreased**, not increased.

B is incorrect because we can't conclude that the sales of the top four firms fell; we can only rightly conclude that their market shares are falling. This is because, the formula of market share is: (Sales of the firm/Sales of the industry)*100. It may be that the market share of these firms is falling due to a rise in the total sales of the industry.

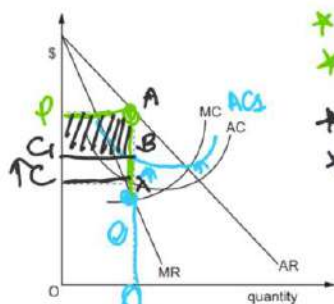
C states that Scotsdale and Welshpool's profits rose — but the table only shows **market share**, not revenue or cost data, so profit conclusions cannot be drawn.

6-D:

Correct answer: D — Price unchanged; Output unchanged; Profit falls

An increase in **fixed costs** raises **average total costs (ATC)**, but it has **no effect on marginal cost (MC)**. Since price and output at the profit-maximizing level of output are determined where $MC = MR$, and this intersection doesn't shift MC, both price and output remain unchanged. However, since ATC has increased at the same output level, **profit per unit falls**, and total profit declines.

6 The diagram shows a profit maximising firm in equilibrium.



What would happen to the firm's price, output and profit if there was an increase in its fixed costs?

	price	output	profit
A	rise	fall	rise
B	rise	fall	fall
C	unchanged	rise	rise
D	unchanged	unchanged	fall

7-C:

The firm charges **different prices for the same service** depending on when the ticket is purchased, without corresponding differences in cost (**the reasons do not depend on cost, but instead depend on the price-elasticity of demand**)

This is **price discrimination**, where the firm is now charging a relatively higher price for the ticket as the departure date of the airplane approaches. This is because the demand for the passengers now wishing to purchase the ticket will be relatively price-inelastic. This is how the firm extracts **consumer surplus** by segmenting buyers based on time sensitivity.

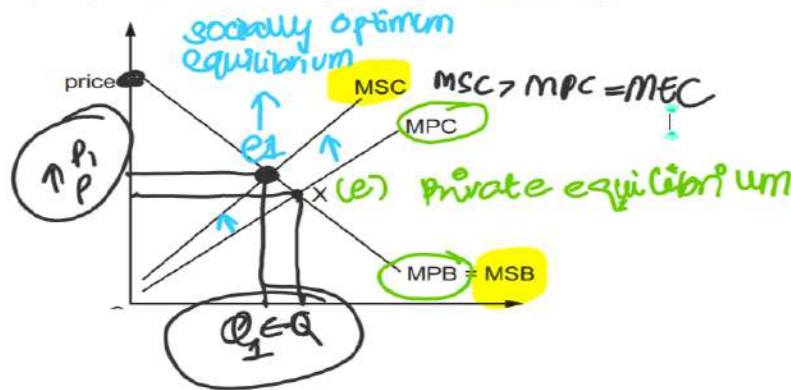
A refers to government-imposed minimum prices, usually above equilibrium to protect producers. This is not the case here.

B refers to aggressive underpricing (**sometimes prices are even illegally set below costs**) to drive out competitors, but prices are rising here, not falling below cost.

D refers to a dominant firm setting prices for the industry; this is typically seen in case of a dominant oligopoly — there's no evidence of this dynamic.

8-A:

The diagram shows the marginal private benefit (MPB), the marginal private cost (MPC), and the marginal social cost (MSC) for firms in an industry. The equilibrium price is point X.



The diagram shows $MSC > MPC$, meaning **negative externality** exists. The private market equilibrium is at point X, where $MPB = MPC$. However, socially optimal equilibrium is at “e1” where $MSB = MSC$, which is at a lower quantity and higher price. To achieve **allocative efficiency**, output must fall and price must rise to reflect the **full social cost** of production.

This explains why key A becomes correct.

9-C:

This is a classic **game theory scenario**, where each firm must decide whether to **cooperate** or **compete** without knowing the other’s decision. Cooperation yields shared profit (\$2000 each), but if one defects and undercuts the other, it can earn more (\$2800) while the other earns less (\$1200). The lack of trust and interdependence creates the prisoner’s dilemma, where **self-interest may lead to worse outcomes** for both firms.

A is incorrect — this is not a monopoly maximizing profits. Instead, it is a dilemma faced by firms in a duopoly.

B relates to conflicts between owners and managers, not rival firms.

D is incorrect because satisficing refers to settling for satisfactory outcomes, but this is not the case here.

10-C: When a smartphone company acquires a **chip design firm**, it secures access to a key input. This is **backward vertical integration**, often used to reduce dependency for inputs, ensure component quality, and reduce production costs.

A is incorrect because horizontal integration involves merging with a **competitor**, not a supplier. **B** refers to lateral integration, which isn’t applicable here as both firms are in the same production chain.

D would be correct if the chip design firm acquired the smartphone manufacturer (that would be **forward** integration)

11-B:

A) This is a valid reason for subsidizing green energy because coal is abundant and cheap, making it the more attractive option in a free market. To make green energy competitive and shift consumption away from polluting fuels, governments may subsidize cleaner alternatives like solar and wind.

B) This is **not** a valid reason to subsidize green energy because, in economic theory, markets are already assumed to be efficient unless there is a clear market failure. Subsidies, by themselves, distort market prices and can lead to overproduction and deadweight loss. Therefore, claiming that subsidies will simply “make markets more efficient” is vague and unsupported unless linked to a specific failure. Subsidies also lead to producers becoming more inefficient.

C) Green energy provides positive externalities like cleaner air, but if these can’t be measured, the market underproduces it. Subsidies help correct this by encouraging more use despite the benefits being hard to quantify.

D) Fossil fuels cause harm like pollution (negative externalities), but if the full cost isn’t known or priced, they remain too cheap. Subsidizing green energy offsets this distortion and promotes cleaner alternatives.

12-D:

In a **monopsony labor market**, the firm has wage-setting power and may employ fewer workers than socially optimal to minimize costs.

When a **minimum wage is introduced** and set below the worker’s MRP, this means that the Marginal Revenue Productivity is still higher than the Marginal Cost of Labor (the cost of hiring an additional labor). As a result, it will still be profitable for the profit-maximizing monopsonist to hire more labor because it can hire workers up to the point where $MRP = MCL$.

This is how it can increase both **wages and employment**.

A is incorrect because intense competition makes firms highly cost-sensitive — a mandated wage rise may force layoffs, not boost jobs.

B is incorrect; if labor costs are a large portion of total costs, raising wages increases overall expenses significantly and may reduce employment significantly because in that case the demand for labor will be elastic.

C is incorrect because a minimum wage may still distort hiring decisions even if it doesn’t reduce profits substantially. It still has the potential to generate unemployment.

13-B: In this question, the marginal revenue productivity curve or the demand for labor curve is shifting towards the left.

When the price of the final product falls, its quantity supplied will contract. In response to a fall in production, there will be a lesser need of workers.

Therefore, the demand for labor, which is a derived demand, will fall. Consequently, there will be a leftward shift in the MRP curve, making key B correct.

Similarly, in event of a rise in the price of the final product, firms will respond to price signals by increasing their production (quantity supplied), thereby requiring more labor to meet the rise in output. This will result in a rightward shift in the MRP curve, making D incorrect.

A is incorrect because wage changes cause **movement along** the MRP curve, not shifts of the curve itself.

C also refers to a wage change, which again affects quantity demanded but does not shift MRP.

14-C:

A profit-maximizing firm employs labor up to the point where **MRP = marginal cost of labor (MCL)**. According to the table, MRP equals MCL (both at 30) at the third worker. Hiring a fourth worker would yield MRP of only 20, which is **less than the marginal cost**. This explains why key C is correct.

A and B are incorrect because in both cases, $MRP > MCL$, meaning the firm could still increase profit by hiring more.

D is incorrect because $MRP < MCL$ for the fourth worker, indicating hiring that worker would reduce profit.

15-B:

A minimum wage **directly targets low income-earners** and benefits them by raising their earnings. This can reduce **income inequality in the short run**, particularly in sectors where pay is low. While long-term structural reforms (like education) help, minimum wages offer an **immediate redistributive effect**.

It is best to target incomes to reduce income disparities.

A) If through imposition of sales tax, the prices of luxury goods are raised, their qty demanded may contract significantly since luxury goods have an elastic demand, and there may not be enough revenues from taxation earned to reduce income disparities.

C is a long-term policy — it improves human capital but has minimal short-run effect on inequality.

D boosts productivity and growth but may not reduce inequality immediately.

16-C:

The characteristics of money describe its essential physical or qualitative properties, while the functions of money explain what money does in an economy. In essence, characteristics define what makes something suitable to be used as money (examples include divisibility, portability etc.), whereas functions describe how that money is actually used (medium of exchange, store of value, standard of deferred payments, unit of account)

A (divisibility) refers to splitting money into smaller units, which is a characteristic of money.
B (durability) is about money's ability to physically last, which is again a characteristic of money
D (store of value) is a function of money but it refers to **saving** for future use, not contracting for future payment.

Agreeing to pay in the future demonstrates money's role as a **standard of deferred payment**. This function allows contracts, credit arrangements, and long-term agreements to be based on a **stable and trusted medium**, making deferred transactions possible. As a result of this, option C becomes correct.

17-C: The **natural rate of unemployment (equilibrium unemployment)** includes **structural and frictional unemployment**, often caused by skills or geographical mismatches. Increasing labor mobility — both **occupational** and **geographical** — helps workers transition more efficiently to available jobs, reducing the mismatch and therefore **lowering the natural rate**. As a result of this, key C, is correct.

A would increase unemployment. Lower education not only reduces employability, but also makes the workers occupationally immobile if they are not multi-skilled or versatile, making them more prone to being structurally unemployed.

B which involves raising income taxes would reduce incentives to work, which will cause the natural rate of unemployment to rise.

D would do the same — higher benefits reduce the urgency to find work, potentially increasing voluntary unemployment. This will also cause the natural rate to rise.

18-A:

Rising productivity indicates that **output per input** is improving — this is **productive efficiency**. Falling unemployment suggests **labor resources** are being more fully used. When the actual moves closer to the potential, this suggests that the economy as a whole is also closer to achieving productive efficiency. Combined, these suggest improved **overall economic efficiency**.

B) If interest rates were to rise, this would dampen Aggregate demand and result in an increase in unemployment.

C) Higher taxation would normally discourage production, not align with falling unemployment.

D) The labor force does not need to necessarily increase for there to be productivity gains.

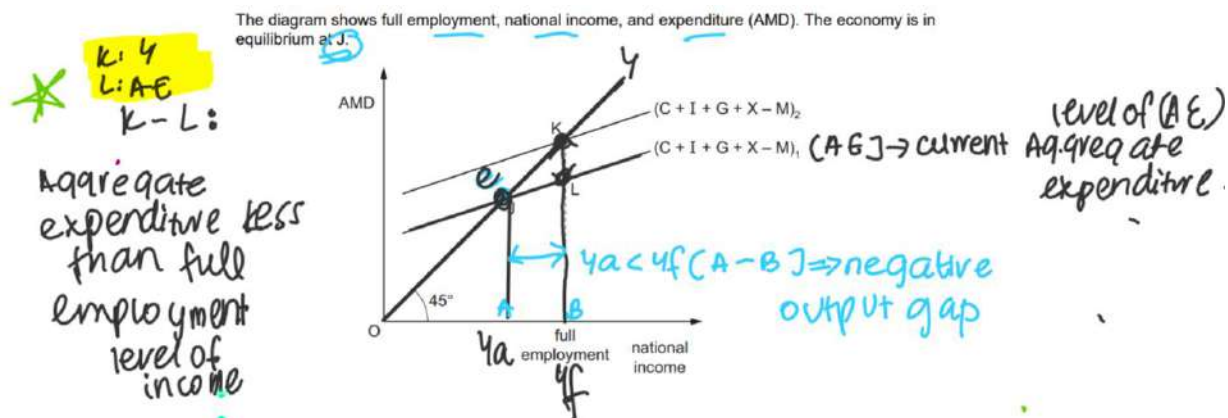
19-A:

The distance KL represents the **gap between full employment level of income** and current **aggregate demand**. Since J lies **below** the full employment level, this is a **deflationary gap** — indicating insufficient Aggregate demand to utilize all productive resources.

B refers to trade imbalance, which cannot be read from this diagram.

C is incorrect — an inflationary gap exists when we are operating above full employment.

D “employment gap” isn’t the correct terminology used in macroeconomic models.



20-A; A — Decrease in tax revenue, increase in government expenditure

A deficiency in aggregate demand leads to **higher cyclical unemployment**. This reduces income and spending, lowering **tax revenues** (especially income and consumption taxes). At the same time, more people become eligible for **unemployment benefits**, increasing **government expenditure**. If tax and benefit rates remain unchanged, this leads to **higher fiscal deficits**.

21-B:

In the short run, an increase in economic growth typically leads to a rise in **real GDP**, which increases firms’ output and income levels. To meet rising demand, firms need to employ more workers, causing a **fall in cyclical unemployment**. Therefore, a successful growth policy tends to raise employment as a derived effect of rising aggregate demand. Hence, B is the most likely macroeconomic aim achieved alongside growth.

A is incorrect because a balance of payments surplus depends on net exports, which may worsen during a period of economic growth as imports rise.

C Growth achieved due to an increase in Aggregate demand can actually cause **demand-pull inflation**, unless matched by supply-side expansion.

D As mentioned above, AD driven growth can result in demand-pull inflation. Inflation can consequently worsen BOT position as the country’s exported goods appear expensive in the foreign market and imported goods appear relatively cheaper. This can reduce the demand for the domestic currency while increasing its supply, thereby causing the ER to depreciate.

22-D:

Lowering income tax reduces the **gap between gross and net income**, making it less worthwhile for individuals to engage in **tax evasion** or avoidance. When marginal tax rates are very high, the

perceived benefit of tax evasion increases. But when tax rates are reduced, this can shift behavior toward **compliance**, even if other effects (e.g. on growth or deficit) are uncertain.

A is not always true — lower income tax may not boost spending and economic activity if there is low confidence in the economy (that is when people may choose to save their extra disposable income instead of spending it)

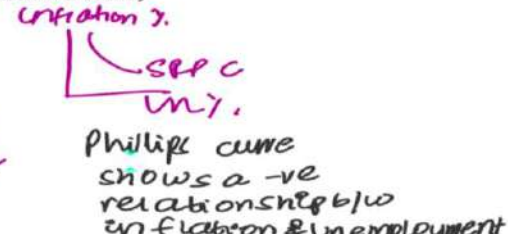
B may not always promote the incentive to work — especially if work hours are rigid or income effects outweigh substitution effects.

C is incorrect because lower income tax often **worsens** the budget deficit.

23-A:

The table shows the percentage of labour force unemployed and the rate of inflation in a country over a five-year period.

year	unemployment %	inflation %
1	3.5	4.0
2	4.1	3.6
3	4.3	2.5
4	4.5	2.2
5	4.8	2.0

inflation %

 -ve
 -ve
 -ve
 -ve
 Phillips curve shows a -ve relationship b/w inflation & unemployment

The Phillips curve shows an **inverse relationship between unemployment and inflation**. In the table, as unemployment steadily rises from 3.5% to 4.8% over five years, inflation falls from 4.0% to 2.0%. This consistent inverse trend matches the original Phillips curve model.

24-C:

The **Gini coefficient** is a quantitative measure of income inequality, ranging from 0 (perfect equality) to 1 (maximum inequality). A lower Gini coefficient signals a more even income distribution. Hence, this is the most precise indicator of improved income equality.

A is incorrect because growth can occur with either rising or falling inequality (**as observed through the Kuznets Curve**) — it's not a distributional measure.

B refers to broader development (health, education, income), not just income distribution.

D is misleading; a lower tax/GDP ratio may reflect **less redistribution**, not more income equality.

25-B:

Expenditure switching policies aims to encourage consumers to switch from imported goods to domestically produced goods.

By placing tariffs on Chinese imports, the US aims to make **foreign goods more expensive**, encouraging consumers to switch toward **domestically produced goods**. This is an example of an expenditure switching policy.

A is incorrect as expenditure-reducing policies aim to dampen overall Aggregate demand. Additionally, monetary policy involves using money supply, interest rates, exchange rate and credit regulations to influence Aggregate demand; this is not happening here. **Therefore, C and D are incorrect as well.**

26-B: The dividend received from abroad represents a **primary income inflow** — part of the **current account**. It's recorded as a **credit** in the current account because money is entering the domestic economy (the US) from a foreign source.

A and C are incorrect — capital account deals with transfers of assets, not income from investments.

D is incorrect because this is clearly an **inflow** (credit) in the current account, not an outflow (debit) in the current account)

27-B:

In a **floating exchange rate system**, currency value is determined by supply and demand. A fall in value is called a **depreciation of the exchange rate**. In a **fixed system**, changes happen by policy decision — a deliberate increase in currency value is a **revaluation**.

28-D:

D – 90% raw materials sourced domestically; income elasticity 1.5

The **best improvement in the current account** comes when imports are low and exports rise. If 90% of inputs are locally sourced, few imports are needed — limiting **import leakage**. A high income-elasticity (1.5) means global demand for electric cars is **very responsive** to income growth, raising exports. Together, these conditions improve the **trade balance**, strengthening the current account.

29-B: When the Lorenz curve moves from L1 to L2, this represents a rise in income inequality as the Lorenz curve moves further away from the line of perfect equality.

Making income tax more progressive directly redistributes income, reducing income inequality. This is **not a valid explanation** for the Lorenz curve moving further away from the line of perfect equality, **making key B correct**.

A) Reducing capital gains tax (progressive tax) benefits the wealthy and widens inequality- it is a valid explanation.

C) When inheritance tax (progressive tax) reduces, income inequality increases- again a valid explanation.

D) Reducing the tax-free threshold would hit low earners hardest, increasing inequality.

30-D:

Globalization enables **international specialization, increased trade**, leading to lower prices and more consumer choices. Over time, these improvements raise **living standards**. This makes key D correct.

A is incorrect — **comparative advantage** remains the basis for trade.

B is false — globalization tends to lengthen supply chains as firms source inputs globally.

C is incorrect — globalization can **increase exposure** to international shocks, as economies become more interdependent.

